

MA 201: Mathematics-III

Course Code	MA 201
Course Name	Mathematics-III
L-T-P-C	2-1-0-3
Intended for	B.Tech
Distribution	Compulsory
Approval	5th Senate (Oct 2013)

Course Contents

- **Linear Algebra:**

Unit I: Vector spaces, Sub Spaces, Linear Dependences and Independences of Vectors, Span, Bases and Dimensions, Direct Sum, Linear Transformations, Linear Variety, Range Space and Rank, Null Space and Nullity, Homomorphism, Matrix of Linear Transformations, Matrix Representation of a linear transformation, Structure of the solutions of the matrix equation $Ax = b$, Change of Bases. (6 hrs)

Unit II: Elementary canonical forms: Triangulation, Diagonalisation of Matrices, Jordan canonical forms and some applications. Normed Linear Space, Inner Product Spaces, Orthogonality, Gram-Schmidt Orthogonalization. (7 hrs)

- **Complex Variable:**

Unit III: Functions of a complex variable: Limit, Continuity, Differentiability, Analytic functions, Cauchy-Riemann equations, Laplace equation. (4 hrs)

Unit IV: Harmonic functions, Harmonic conjugates. Complex logarithm function, Branches and Branch cuts of multiple valued functions. (4 hrs) **Unit V:** Complex integration, Cauchy's integral theorem, Cauchy's integral formula. Liouville's Theorem and Maximum-Modulus theorem, Power series and convergence, Taylor series and Laurent series. (4 hrs)

Unit VI: Zeros, Singularities and its classifications, Residues, Rouches theorem, Argument principle, Residue theorem and its applications to evaluating real integrals and improper integrals. Conformal mappings, Mobius transformation. (5 hrs)

Text Books

1. G.Strang, **Linear Algebra and its Applications**, 4th Edition, Thomson, 2006.
2. R.V.Churchill and J.W.Brown, **Complex Variables and Applications**, McGraw-Hill, 2008.
3. H.Anton, **Elementary Linear Algebra with Applications**, 9th Edition, John Wiley, 2004.

References

1. K. Hoffman and R. Kunze, **Linear Algebra**, Prentice Hall, 2008.
2. J.M.Howie, **Complex Analysis**, Springer-Verlag, 2003.
3. M.J.Ablowitz and A.S.Fokas, **Complex Variables- Introduction and Applications**, Cambridge University Press, 2003.
4. E. G. Phillips, **Functions of a Complex Variable: with Applications**, Barnes & Noble Books, 1973.
5. S.Kumaresan, **Linear Algebra – A Geometric Approach**, Prentice Hall of India, 2004.